

Ques. WHAT ARE COLOUR SENSORS AND WHICH SENSORS ARE MOST PREFERABLE?

Pamarthi Kanakaraja

Ans. The colour sensor product family provides red, green, blue and clear (RGBC) light sensing for precise colour measurement, determination and discrimination. A sync input allows for greater accuracy by enabling the colour sensing to be synchronised with an external event. For example, in LED solid-state lighting, the sync input enables the colour reading to be synchronised with the PWM signal and illumination of the LEDs. Colour sensors are available with a built-in infrared (IR) blocking filter for accurate colour measurement by minimising the effects due to unwanted IR light sources in sunlight and other light sources.

Colour sensors from ams AG are available in TCS series as RGB light-to-voltage converters (TCS3103, TCS3104), programmable colour light-to-frequency converters (TCS3200, TCS3210), with sync input for colour-sensing synchronisation and in-package IR-blocking filter (TCS3404, TCS3414), colour light-to-digital converter I2C (TCS34715), colour light-to-digital converter I2C and IR filter, etc.

EMX has a number of colour sensors available in ColorMax-1000 series. A sensor can recognise 15 colours and RGB intensity. Details are available on their website www.emxinc.com/color_sensor.html

Q2. WHAT IS THE DIFFERENCE BETWEEN SSP AND SPI IN ARM PROCESSOR?

Pamarthi Kanakaraja

A2. Synchronous serial port (SSP)

module is a serial interface useful for communicating with other peripherals or microcontroller devices. SSP is a controller that supports the serial peripheral interface (SPI), 4-wire synchronous serial interface (SSI) and micro-wire serial buses. It uses a master-slave paradigm to communicate with peripheral devices. These peripheral devices may be serial EEPROMs, shift registers, display drivers or A/D converters. An SSP module can operate in one of two modes:

1. Serial peripheral interface (SPI)
2. Inter-integrated circuit (I2C)

SPI is a hardware/firmware communications protocol developed by Motorola and later adopted by others in the industry. It is a serial communications interface used by many microcontrollers that enables the controllers and peripheral devices to communicate with each other to transfer data at high speeds.

The SPI bus, which operates at full duplex (means, signals carrying data can go in both directions simultaneously), is a synchronous type data link set up with a master/slave interface. It is usually used only on the PCB.

SSP peripheral is a multi-protocol peripheral, which is also able to function as an SPI interface. Since the SSP interface is newer and more versatile, the SPI interface is not recommended for new designs.

Q3. HOW TO SYNCHRONISE RTC WITH A PC VIA ARDUINO? PLEASE PROVIDE THE LINK FOR THE SAME.

Maheesh Kumar

A3. To use a real-time clock (RTC) with an Arduino board, first make Arduino ready by uploading the program to the board. The program will receive the time from the PC in a special

format and write it on the RTC. To see the current time of the RTC you need to send a command using the serial port.

To send the time from PC, you just need to run a batch file. Change the COM number in the batch file. Do not forget, your PC's serial port should be free. If serial communication of your Arduino IDE is open, close it first and then run the batch file.

Now, if you send command on the Arduino's serial port, you will see the same time as your PC. For your reference please follow the link www.instructables.com/id/Synchronise-DS3221-RTC-with-PC-via-Arduino/

Q4. CAN I USE ATMEGA16 AND ATMEGA8 CHIPS WITH AN ARDUINO UNO BOARD THAT IS TO BE PROGRAMMED?

Vignesh Vicky

A4. You cannot use Arduino Uno board directly with ATmega16 and ATmega8. To work with ATmega16 and ATmega8 using Arduino IDE one has to use Arduino-compatible boards. Among the boards that are fully or almost fully compatible with both the Arduino hardware and software, including being able to accept 'shield' daughterboards, is the Elektor Platino Universal AVR board (compatible with ATmega8, ATmega16, ATmega32, ATmega88, ATmega164, ATmega168, ATmega324, ATmega328, ATmega644 and ATmega1284), which supports 28-pin and 40-pin AVR devices. The board features multiple footprints for user interface elements like LCDs, pushbuttons, rotary encoders, LEDs and buzzer, and is supported by an extensive library. Bootloaders are available for all supported processors.

Answers compiled by EFY senior application engineer, Nidhi Kathuria. Letters and questions for publication may be addressed to Editor, Electronics For You, D-97/1, Okhla Industrial Area, Phase 1, New Delhi 110020 (e-mail: editor@efy.in) and should include name and address of the sender